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Vascular Diseases

JASMINE ZHENG, ESMAEEL R. DADASHZADEH, AND RYAN MICHAEL COBB

SUMMARY PEARLS

- The CEAP clinical score is used as the standard for reporting venous disease, and the VEINES-QOL is used widely to assess the effect of chronic venous disease on quality of life.
- Continuous wave Doppler, duplex ultrasound, computed tomography, magnetic resonance venography, contrast venography, and intravascular ultrasound are imaging modalities used to evaluate the intravascular system.
- Patients with moderate to severe postthrombotic syndrome should be referred to a venous specialist for endovascular treatment options while maintaining clinical care with their outpatient team for symptom management and wound care.
- Phlegmasia alba dolens is a rare complication of deep vein thrombosis characterized by rapid and massive edema, pain, and skin blanching, and this requires urgent treatment.
- May-Thurner syndrome is due to compression of the left iliac vein by the right common iliac artery and can cause isolated left lower extremity swelling that can mimic lymphedema.
- The functions of the lymphatic system include transporting fluids from the body's intercellular spaces back into the venous system, supporting immune functions, and absorbing and moving fatty acids from the gastrointestinal system.
- Lymphedema is the abnormal buildup of water and proteins in the subcutaneous tissues due to abnormality in the lymphatic system, and it is estimated to affect more than 200 million people worldwide.
- Central lymphatic diseases are due to two main mechanisms: direct trauma to lymphatic vessels and occlusion of the thoracic duct that leads to leaky collateral vessels.
- Lymphoscintigraphy, magnetic resonance lymphangiography, indocyanine green lymphography, and ultrasound are imaging modalities used to view the lymphatic system.
- The gold standard of care in treatment of lymphedema is complete decongestive therapy.

Introduction

Vascular and lymphatic diseases are commonly seen in the rehabilitation population. The physiatrist is often the first physician to evaluate patients with difficulty in ambulation secondary to an underlying and yet to be diagnosed vascular pathology. Patients may present with pain, wounds, and/or edema that, if untreated, can lead to problems with range of motion, mobility, and the ability to perform activities of daily living and instrumental activities of daily living. The purpose of this chapter is not only to guide the physiatrist with the initial assessment and diagnostic testing of patients with suspected vascular disease but also to advise on the appropriate timing and urgency of treatments and referrals.

Arterial Disease

Peripheral arterial disease (PAD) is defined as the presence of atherosclerotic occlusive disease of the lower extremity arterial vasculature. Given that many patients with PAD are asymptomatic, its true incidence is difficult to ascertain. Estimations based on available screening data conservatively predict that nearly 200 million people worldwide and 12 million Americans live with PAD.[21] Those with symptoms present with intermittent claudication, defined as cramping of the muscle compartment one anatomic level below the level of the arterial stenosis. For instance, those with iliac artery disease present with hip/buttock claudication, and those with superficial femoral artery disease present with calf claudication. Patients describe the onset of symptoms at a fixed, predictable amount of ambulation, which is always relieved by rest. Given that PAD is a polyvascular disease, the overall 5-year mortality is 15% to 30%, three-quarters of which are due to cardiovascular causes.[67] Approximately 1% to 2% of patients with PAD progress to chronic limb-threatening ischemia (CLTI) over a 5-year period. CLTI is defined as PAD with the presence of wounds or ischemic pain even at rest, often localized on the dorsum of the foot. This is a morbid condition that requires early evaluation because it carries a 25% 1-year mortality and a 25% 1-year incidence of limb loss.[47] Given its status as one of the most underdiagnosed vascular pathologies, prompt recognition and management of patients with CLTI can be both limb and life saving.

Diagnosis of Peripheral Arterial Disease

History and Physical Examination

When evaluating a patient with any lower extremity or ambulation complaints, there are several elements of the history and physical examination that can elevate PAD in the differential diagnosis. Age and cigarette smoking are the greatest associated risk

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• **Fig. 26.1** Note the acute demarcation of the limb hairline in the distal calf, seen in a patient with peripheral arterial disease.

factors. Even in the presence of smoking, PAD is rare in those under the age of 40 years, but the incidence steadily rises every decade thereafter, approaching 25% for octogenarians.[47] Additional risk factors in decreasing association include diabetes, hypertension, dyslipidemia, and obesity.[15] Of patients with PAD, 60% are asymptomatic, making the diagnosis challenging. Even in the absence of symptoms, patients with PAD carry the same elevated risk of cardiovascular mortality and can be offered a survival benefit with proper risk factor modifications (see Management, later). Aside from a review of the associated risk factors, the physical examination may also assist in diagnosis. Patients with PAD may demonstrate atrophied lower extremity musculature, cool skin, and an abrupt demarcation of the limb's hairline (Fig. 26.1). The pedal pulse exam is also a vital tool. In the asymptomatic patient, the absence of a palpable posterior tibial artery and/or dorsalis pedal artery pulse is a poor predictor of PAD and should not raise any alarms. The same can be said with regard to absent Doppler signals in a patient with no acute symptomatic complaints. In fact, the presence of a palpable pedal pulse strongly predicts against PAD in the asymptomatic patient.[34]

In patients presenting with intermittent claudication (25% of the PAD population), the pulse exam can not only assist in the diagnosis but also help determine the anatomic level of disease. While there are many etiologies of lower extremity pain, claudication is specific to PAD. Patients with claudication describe reproducible muscle pain with exertion that resolves with rest. An absent femoral artery pulse with hip or buttock claudication is associated with iliac artery atherosclerotic disease, and an absent pedal pulse with calf claudication likely points to the superficial femoral artery as the culprit. Patients with isolated tibial disease may also have a pedal pulse irregularity; however, they are unlikely to have any claudication symptoms.[16]

A thorough inspection of both lower extremities of any patient with symptomatic PAD is warranted because the identification of any wound present for more than 2 weeks (gangrene or ulceration) upgrades their diagnosis from PAD to CLTI, which encompasses 1% to 5% of the PAD patient population. These wounds are often located in the most distal arterial beds, such as the digits, and may be subtle in their initial appearance (Fig. 26.2). However, any wound even more proximal is still an urgent matter in any patient with arterial insufficiency because the risk for limb loss remains elevated without intervention. CLTI can also be diagnosed in the absence of wounds when the patient endorses foot pain at rest. On examination, one may note dependent rubor of the foot, which appears as erythema (often confused for cellulitis) that improves with leg elevation.[14]

Noninvasive Arterial Studies

Ankle-Brachial Index

When suspecting PAD or CLTI, there are several noninvasive vascular studies the physiatrist can order prior to or in concordance with vascular surgery consultation, all of which impart no risk to the patient. The simplest is the ankle-brachial index (ABI). This measurement can be obtained by the vascular lab or at the bedside by any provider with a portable pencil arterial Doppler probe and a blood pressure cuff.

1. Place the cuff above the ankle of the extremity under examination.
2. Place the Doppler probe over the dorsalis pedis artery where the audible signal is at its highest amplitude.
3. Maintain still position of the Doppler probe and inflate the cuff until the Doppler signal is no longer audible.
4. Release the cuff pressure, then note the pressure reading at which the Doppler signal returns.
5. Repeat for the posterior tibialis artery. Select the higher of the two measurements.
6. Take a brachial cuff blood pressure measurement of both arms. Select the higher of the two measurements.
7. Divide the higher ankle measurement by the higher brachial measurement to obtain the ABI.

An ABI of 0.91 to 1.29 is considered normal. Patients with mild to moderate PAD often have an ABI of 0.41 to 0.90, and those with severe PAD and possibly CLTI often have an ABI of 0.00 to 0.40. Patients with underlying diabetes or end-stage renal disease may have noncompressible vessels due to medial calcific sclerosis of their tibial vessels, resulting in falsely elevated ABIs greater than 1.30, a result that offers no diagnostic value.[72]

Segmental Pressure Measurements

Many vascular labs across the country obtain segmental pressures along with ABIs when an arterial Doppler study is ordered. Cuffs are placed at the upper thigh, lower thigh, calf, and ankle, and like an ABI measurement, a Doppler probe is placed on a pedal artery while each cuff is sequentially inflated and deflated. A pressure drop of 20 mm Hg between any two levels indicates significant disease.[3] A segmental pressure at the level of the toe can also be obtained. This is especially useful in patients with noncompressible ABIs or with wounds because the digital microcirculation is less affected by medial calcific sclerosis. The toe-brachial index is obtained in a similar manner to the ABI, and a value of 0.7 or greater is considered normal. An isolated toe pressure measurement can help assess the potential for wound healing. A pressure less than 30 mm Hg is a poor predictor of wound healing, and a pressure greater than 60 mm Hg is likely adequate.[57]

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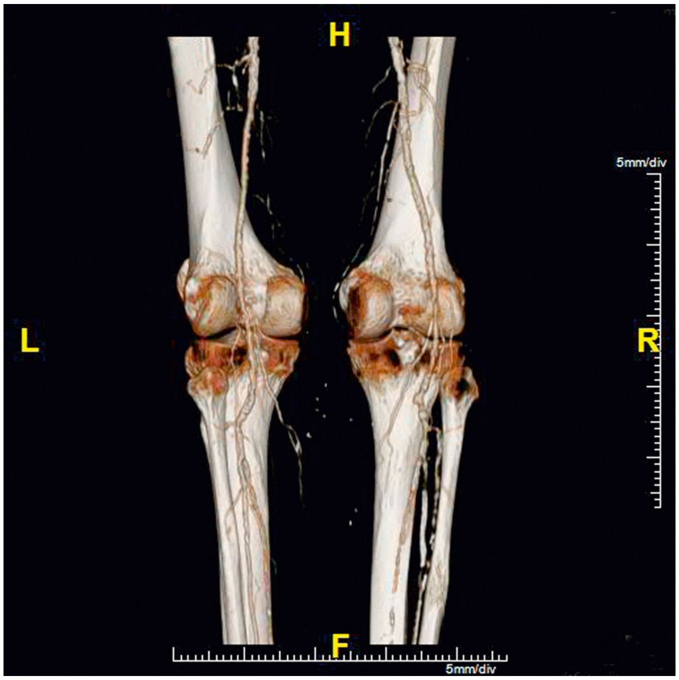
• **Fig. 26.2** (A, B) Subtle ulcerations of the great toenail beds in patients with chronic limb-threatening ischemia.

Exercise Testing

The physiatrist may often encounter a patient in the early stages of PAD, where symptoms of claudication are endorsed, but resting ABI studies return as normal. Although none should fault invasive imaging or vascular surgery consultation as this point, an alternative noninvasive test to consider is exercise ABI. An arterial stenosis may be severe enough to produce symptoms but not yet result in a drop in resting ABI. Assuming that the upper extremity vasculature is free of disease, exercise would cause a predictable increase in the brachial cuff measurement but a diminished increase in the ankle cuff measurement due to a proximal stenosis. First, resting ABIs are obtained after 20 minutes of lying supine. Next, the patient is asked to walk on an elevated treadmill for 5 minutes. Afterwards, ABIs are obtained every 2 minutes until the prior resting ABI is achieved. A patient that can complete the test with no drop in ABIs can be ruled out for clinically significant PAD. As the severity of PAD increases, the time postexercise to normalization of ABI increases as well. An absolute ankle pressure postexercise 20 mm Hg lower than the resting ankle pressure is also indicative of mild disease, and an isolated ankle pressure less than 50 mm Hg postexercise is indicative of severe disease.[1]

Arterial Duplex

An arterial duplex study uses B-mode ultrasound imaging to directly visualize the vasculature and obtain velocity measurements. This has little utility in patients with no prior vascular intervention. Rather, it serves as the optimal means of surveillance in patients with existing stents and bypass grafts.



• **Fig. 26.3** Computed tomography angiography of a patient with heavily calcified atheromatous disease, including occlusion of the bilateral anterior tibial and posterior tibial arteries. Dominant runoff via diseased peroneal arteries with probable occlusion of the proximal right peroneal artery.

Invasive Imaging Studies

Computed Tomography Angiography

The aforementioned noninvasive studies impart little to no risk to the patient and would be appropriate to order prior to vascular surgery consultation. Invasive imaging, dominated by computed tomography angiography (CTA) (Fig. 26.3), has a minimal risk of contrast-induced kidney injury, although recent studies show this risk to be overstated in the past.[12] Patient history, physical examination, and noninvasive studies are often more than sufficient to

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secure the diagnosis of PAD. CTA serves a greater role for operative planning. If a CTA is ordered, it should always include the abdomen and pelvis along with bilateral lower extremity runoffs down to the toes to avoid repeat imaging.

Management

With a 15% to 30% 5-year overall mortality, once PAD is diagnosed, the focus of the patient's care should be on cardiovascular prevention medications, with antiplatelet and statin therapy being the most important. Antiplatelet therapy has demonstrated up to 30% risk reduction of stroke, nonfatal myocardial infarction, and vascular death in patients with PAD.[5] While some studies favor clopidogrel at 75 mg daily, PAD guidelines support the use of aspirin at 81 mg daily as well.[11] Statins have demonstrated reduction in amputation rates, cardiovascular events, and improvement in symptoms in patients with claudication. Guidelines recommend that patients younger than 75 years of age be placed on a high-intensity statin (atorvastatin 40–80 mg or rosuvastatin 20–40 mg).[71] Patients with a new diagnosis of CLTI will undoubtedly undergo vascular surgical intervention in the weeks following evaluation. Those with PAD, unless the symptoms are lifestyle limiting, will most often undergo nonoperative therapy, consisting of risk factor modification, aspirin and statin therapy, and exercise therapy to induce collateralization. For the rehabilitation professional treating a patient with incidental PAD, there are no restrictions on movement or exercise. In fact, more physical activity is usually encouraged. The only exception is compression therapy. In patients with PAD and no prior vascular intervention, a good rule of thumb is to avoid compression higher than the patient's toe pressure. In patients with a prior bypass graft, all compression therapy should be withheld prior to consultation with their vascular surgeon.[48]

Rehabilitation

Patients with PAD should be instructed to always wear protective footwear and to monitor their extremities carefully for redness or skin breakdown. Temperature extremes should be avoided. The feet should be washed carefully with mild soap and warm water. Drying is best performed by blotting or patting with a soft, clean towel. Rubbing should be avoided because it may injure the skin. The skin between the toes should be carefully dried to avoid maceration. Emollients without preservatives or perfume should be used to prevent cracking of the skin (avoid between the toes). Proper footwear, which does not produce areas of point pressure, should be used. Whenever new shoes are purchased, the patient should gradually, over a period of 1 week, wear-in the shoes to ensure that there are no areas of point pressure with the new footwear. Warm outer footwear should be used in the winter to protect against thermal injury. Decreased activity secondary to symptomatic lower extremity arterial occlusive disease can result in deconditioning, which further contributes to disease impairment. Deconditioning may also be iatrogenic because of a prolonged period of limited mobility to avoid trauma to ischemic wounds.

Regular lower extremity exercise in the form of a structured or supervised walking program is critical for patients with PAD. Ambulation can help to develop collateral blood flow and in time may lead to resolution or improvement of intermittent claudication. A minimum of 30 minutes of moderate activity at least three times per week is beneficial.[64] Regular training has been shown to improve oxygen extraction from blood, muscle enzyme activity, and hemorheology.[64] Regular exercise training also produces a reduction in the inflammatory markers associated with endothelial damage.[31] Evidence suggests that patients following an exercise regimen improve both their claudication distance and cardiovascular risk profile. Exercise improves maximal walking ability by an average of 150%.[39] Remarkably, walking capacity increased after 6 months following supervised exercise training cessation, suggesting an ongoing benefit of the intervention.[58]

Venous Disease

Venous disease includes acute or chronic occlusion or dysfunction of the systemic venous or pulmonary arterial system. Chronic venous disease is a spectrum of diseases and disorders of the limbs with spider veins and varicosities on one end of the spectrum and edema, skin changes (stasis, hyperpigmentation), and ulceration on the other end (Fig. 26.4). The cause is either nonthrombotic (primary valvular incompetence or venous hypertension due to central venous stenosis) or postphlebitic/postthrombotic syndrome (PTS) secondary to previous deep venous thrombosis (DVT).

Venous Thromboembolism

The disease burden from venous thromboembolism (VTE) is major. Each year there are an estimated 900,000 patients with clinical evidence of VTE in the United States resulting in an estimated 300,000 deaths from pulmonary embolism.[29] When a patient presents with the possibility of a DVT, predisposing risk factors, such as prolonged immobilization during car or plane trips, use of estrogen, previous DVT, family history of thrombosis, or genetic factors, should be elicited. Hypercoagulable states, such as those associated with cancer or inherited coagulopathies and those caused by vessel wall injury from surgery or local trauma, are major predisposing factors for thrombosis. Approximately 30% of surviving cases develop recurrent VTE within 10 years.[29] Independent predictors for recurrence include increasing age, obesity, malignant neoplasm, and extremity paresis.[29] Approximately 28% of patients develop venous stasis syndrome within 20 years.[29] To reduce VTE incidence, improve survival, and prevent recurrence and complications, patients with these

• **Fig. 26.4** Recurrent perimalleolar wound in a 67-year-old female with chronic venous insufficiency with varicose veins and stasis dermatitis.

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characteristics should receive appropriate prophylaxis.[30] Thrombosis may occur anywhere but most commonly involves the deep veins of the leg. Once a thrombosis forms, several events may occur: (1) The thrombosis may propagate, (2) it may embolize, (3) it may be removed by fibrinolytic activity, and (4) it may undergo organization (including recanalization and retraction). An initial inflammatory response leads to fibroblast and capillary ingrowth, which helps to stabilize the thrombus. Organization occurs over weeks to months as the thrombus becomes incorporated into the vessel wall. Once luminal flow is disturbed, antegrade and retrograde propagation of the thrombus may also be promoted by hemodynamic factors. The competing process of recanalization and recurrent thrombus determines the extent of acute DVT and its sequelae. Venous thrombi rarely lyse completely unless they are subjected to pharmacologic lysis, pharmacomechanical thrombectomy, or mechanical thrombectomy.

The natural history and clinical consequences of a lower extremity venous thrombosis depend on the site of thrombus. Small calf vein thrombi frequently occur, especially in postoperative patients. Of those that are asymptomatic, 50% lyse spontaneously and are unlikely to proceed to deep vein incompetence. A calf thrombosis may remain asymptomatic; the thrombosis often dissolves without sequelae by natural fibrinolysis. However, 5% to 20% of calf vein thrombi propagate proximally. If extension occurs into the popliteal or more proximal veins, the chance of pulmonary embolism increases from less than 5% to approximately 50%.[60]

Multiple societal guidelines outline the appropriate treatment of acute DVT in the lower extremities.[63] While no absolute consensus has been met, treatment strategies are in practice to minimize risk of VTE and morbidity related to PTS and chronic venous insufficiency (CVI). Morbidity related to PTS is measured using the validated Villalta score (Fig. 26.5), which categorizes patients to mild, moderate, or severe PTS based on self-reported symptoms and clinical exam findings.[69] Keeping these clinical goals in mind, it is recommended to clinically follow isolated calf vein DVTs, while treatment of femoral popliteal DVT incorporates the use of anticoagulation. DVT involving the iliac and femoral veins requires a nuanced approach with consultation of venous specialists to determine if thrombolysis or thrombectomy would be beneficial.[2] It is common for patients with iliofemoral DVT to undergo endovascular intervention (thrombolysis or thrombectomy) to reduce the risk of developing recurrent DVT and/or moderate to severe PTS. Patients with moderate to severe PTS and residual chronic DVT are best referred to a venous specialist to explore endovascular options while maintaining clinical care with their outpatient team who provides symptom management and wound care.

Upper extremity DVT is most often the result of intravenous catheters causing trauma to the endothelium.[20,46,68] A much rarer occurrence is venous thoracic outlet syndrome (Paget-Schroetter syndrome), which is thrombosis due to compression of the subclavian vein at the costoclavicular junction.[32]

Villalta Score for Postthrombotic Syndrome

Symptoms/Clinical signs	None	Mild	Moderate	Severe
Symptoms				
Pain	0	1	2	3
Cramps	0	1	2	3
Heaviness	0	1	2	3
Paresthesia	0	1	2	3
Pruritus	0	1	2	3
Clinical signs				
Pretibial edema	0	1	2	3
Skin induration	0	1	2	3
Hyperpigmentation	0	1	2	3
Redness	0	1	2	3
Venous ectasia	0	1	2	3
Pain on calf compression	0	1	2	3
Venous Ulcer	Absent			Present

• **Fig. 26.5** Total Villalta score is the sum of the columns denoting postthrombotic syndrome (PTS) classifications as follow: Mild = 5–9; Moderate = 10–14; Severe = >14. Venous ulceration indicates severe PTS and automatically confers a score of ≥15.

Phlegmasia Alba Dolens and Phlegmasia Cerulea Dolens

Phlegmasia alba or cerulea dolens are rare complications of DVT. Phlegmasia alba dolens is characterized by rapid and massive edema, severe pain, and blanching of the skin due to large clot burden of a proximal DVT. The DVT obstructs the venous outflow overcoming the pressure of the arterial inflow resulting in blanching or pale-appearing lower extremity. It can progress to the limb-threatening complication of phlegmasia cerulea dolens, which occurs when the limb experiences cyanosis and turns blue or purple (Fig. 26.6A).[61] Phlegmasia cerulea dolens most commonly occurs with proximal iliofemoral obstruction with extensive distal thrombus of deep and superficial veins. In phlegmasia, the arterial pulses may not be palpable, although anatomically the arteries are patent. In severe cases, gangrene, which may necessitate amputation, occurs.[61] Urgent treatment, including placing a caval filter, heparinization, and thrombectomy or thrombolysis, if possible, is essential to minimize loss of life or limb. Mechanical and pharmacomechanical (combination of mechanical thrombectomy and thrombolytic agents) thrombectomy by endovascular specialists for limb-threatening phlegmasia is common. Phlegmasia typically involves extensive DVT of the central veins, including the IVC, common and external iliac, and common femoral veins. Thrombectomy clears the central occlusion, and patients can respond rapidly, sometimes within hours, after adequate venous outflow is restored (see Fig. 26.6B). There are many devices currently in use that focus on rapid removal of acute and chronic thrombus via venous vascular sheaths with minimal blood loss and suitable safety profiles. These devices have been added to the toolbox for rapid treatment of patients presenting with phlegmasia.

Nonthrombotic Iliac Vein Lesions

Nonthrombotic iliac vein lesions (NIVLs) are the result of extrinsic compression or narrowing of the common or external iliac veins. May-Thurner syndrome is a type of NIVL defined as isolated left lower extremity swelling caused by compression of the left iliac vein by the right common iliac artery as the left common iliac vein passes anterior to the spine. Treatment of NIVLs has historically involved anticoagulation therapy. Advances in interventional management have allowed relief of the associated mechanical compression by open surgical or endovascular repair. If this diagnosis is suspected, referral to a venous specialist is recommended. Appropriate workup can involve cross-sectional imaging,

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• **Fig. 26.6** (A) Phlegmasia cerulea dolens of the left lower extremity secondary to acute deep venous thrombosis of the left common iliac vein through the left popliteal vein is a vascular emergency. (B) Postoperative day 1 after mechanical thrombectomy of the left common iliac through popliteal veins showing near complete resolution of left lower extremity swelling and return to normal skin color.

catheter-directed venography, and intravascular ultrasound (IVUS). Extrinsic compression can be seen on cross-sectional imaging and venography, and intravascular scarring called synechiae can be noted on venography and IVUS.[22] Common endovascular treatment involves balloon angioplasty to disrupt synechiae, if present, and venous stent placement to ensure long-term venous patency without restenosis.

Chronic Venous Insufficiency

Chronic venous disease is an important cause of discomfort and disability and is present in a significant percentage of the population worldwide. A clinical-etiologic-anatomic- pathophysiologic (CEAP) score has been developed as a standard for reporting venous disease.[6] Additional metrics, including the VEINES-QOL, have been validated and are widely used by venous specialists to assess the affect chronic venous disease has on patients' quality of life.[37] Many factors can result in the development of venous insufficiency, including heredity, local trauma, thrombosis, and intrinsic defects in the veins or valves themselves. Venous flow is based on a force that pushes the blood proximally, an adequate outflow, and the presence of competent valves limiting reflux. Any disruption of these components results in chronic venous hypertension. Normally, the pressure in the leg veins is equal to the hydrostatic pressure from a vertical column of blood extending to the right atrium of the heart. At the ankle level, the hydrostatic pressure is approximately 90 mm Hg.[43] The pumping action of the calf muscles during exercise reduces this venous pressure by two-thirds. Even slight muscular movements during normal standing will lower the pressure.[43] Patients with venous insufficiency might fail to reduce ankle pressure or show a rapid return of venous pressure to resting levels at the end of the exercise. The time required for the ankle vein pressure to return to resting levels after exercise is an indicator of the degree of reflux in the limb. Elevated ambulatory venous pressure is associated with an increased incidence of ulceration.[49] When ambulatory, venous ankle pressure is less than 30 mm Hg, and the incidence of ulceration is close to 0%. The incidence of ulceration increases linearly when ankle pressure is greater than 30 mm Hg, reaching 100% when the ambulatory venous pressure is greater than 90 mm Hg.[49] The superficial leg veins normally carry 10% to 15% of the venous return. Incompetent valves in the superficial veins alone rarely cause serious venous hypertension, although 10% of patients with venous ulcers have superficial venous incompetence alone.

Primary deep venous reflux is more often observed in patients who are obese and is more persistent following eradication of the superficial reflux than for patients who are of normal weight.[70] One-fourth of patients with a first, unprovoked, unilateral, proximal DVT who were free of clinically significant primary venous insufficiency showed PTS 5 to 7 months after the index event. Obesity, mild contralateral venous ectasia, poor international normalized ratio control, and the presence of ultrasonographic residual venous obstruction significantly increased the risk of PTS in this population.[23] Venous insufficiency may present up to 5 to 10 years after resolution of the acute episode of thrombosis. Postphlebitic damage in the deep veins is an important cause of CVI. Advanced venous insufficiency develops when the valves in the perforator or deep veins are also incompetent. Ultimately, venous hypertension is the result of valve damage and retrograde venous blood flow to the superficial veins.

Venous Evaluation

Continuous Wave Doppler

Continuous wave Doppler (described earlier) is portable, inexpensive, and used clinically as a screening tool to test the integrity

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of the venous system. This method can identify the presence of venous obstruction or incompetence, quantify the severity of the venous disease, and localize these abnormalities to a particular segment of the limb. The venous flow signal is obtained at several sites in the limb. The patency, spontaneous flow, phasicity, augmentation, competency, and pulsatility of the venous flow are determined and graded. Normal venous flow is spontaneous and phasic with respiration. Obstruction of a vein is characterized by the absence of normal spontaneous venous flow or by the loss of phasic variation with respiration. If the Doppler probe is placed directly over an obstruction, there is absence of spontaneous flow. If the probe is placed below the site of obstruction, there is a loss of phasic change in the venous flow with respiration (a monophasic low-frequency signal). Several maneuvers (deep breathing, Valsalva, and distal compression of the calf or forearm) can produce augmentation of venous flow and highlight valve competency. Continuous wave Doppler provides subjective information. If positive for obstruction, findings should be followed by an objective test. Because of this and other limitations, continuous wave Doppler ultrasound has largely been replaced by venous duplex scanning for the diagnosis of DVT (combines Doppler principles with real-time B-mode and color-flow ultrasound imaging).

Duplex Ultrasound

Duplex scanning has become the method of choice for testing veins of the superficial, deep, and perforating systems. Duplex ultrasound directly visualizes and locates intraluminal obstruction, assesses the characteristics of venous flow distal to the inguinal ligament, identifies the presence of collateral veins around an obstructed venous segment, permits direct detection of valvular reflux, allows visualization of specific venous valves and valve leaflet motion, quantifies the degree of incompetence, locates and assesses veins before harvest for bypass procedures, evaluates venous perforator incompetence, and evaluates conditions that may mimic venous disease.

Vascular Disease Computed Tomography and Magnetic Resonance Venography

Early venous disease rarely requires advanced imaging studies other than duplex ultrasonography. Computed tomography and magnetic resonance imaging (MRI) have progressed tremendously in the past decade. Both modalities are suitable to identify pelvic or iliac venous obstruction in patients with lower limb varicosities when proximal obstruction or iliac vein compression (NIVL) is suspected. Gadolinium-enhanced MRI is especially useful in evaluating patients with vascular malformations, including those with congenital varicose veins.[38]

Contrast Venography

Lower extremity contrast venography remains a powerful, but decreasingly used, tool in the initial diagnosis of both acute and chronic DVT. With advances in duplex ultrasonography, venography has been largely replaced by duplex scanning to evaluate patients with suspected deep venous obstruction or incompetence. In chronic venous disease, ascending venography demonstrates the location and extent of postthrombotic disease (Fig. 26.7) as manifested by occlusion, venous recanalization, collateral channels, and superficial varicosities. Ascending venography may also help

• **Fig. 26.7** (A) Venogram showing severe narrowing of the femoral at the upper thigh level, and drainage preferentially via a large incompetent perforator to the great saphenous vein at the midthigh level. (B) The external iliac veins are occluded bilaterally with extensive pelvic collaterals draining via the paraspinal venous plexus.

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with planning of endovascular and open surgical procedures, such as iliac and inferior vena cava recanalization and venous bypass graft. Ascending contrast venography is performed primarily in patients with significant chronic deep venous occlusive disease who are candidates for endovascular treatment with stents, surgical venous bypass, venous valve repair, or valve transplantation. Descending venography is used in concert with ascending venography to distinguish primary valvular incompetence from thrombotic disease. Descending venography identifies the level of deep vein reflux and evaluates the morphology of the venous valve.

Intravascular Ultrasound

IVUS can be an adjunct to catheter-directed contrast venography. As the name entails, IVUS gives a real-time view of the intravascular and perivascular tissues. It is used to assess for acute and chronic DVT, synechiae, venous compression, and stenosis. IVUS allows endovascular mapping and planning to address both thrombotic and nonthrombotic lesions that result in reduced venous return and venous insufficiency. IVUS allows for accurate mapping for thrombectomy and venous stent sizing prior to placement. The use of IVUS has helped venous specialists to predict clinical outcomes after stent placement, in particular, in patients with NIVLs.[22]

D-Dimer

Fibrin D-dimer is the final product of the plasmin-mediated degradation of cross-linked fibrin. The plasma concentration of D-dimer depends on fibrin generation and subsequent degradation by the fibrinolytic system.[52] D-dimer levels have a high sensitivity for patients with acute venous thrombosis.[56] However, the specificity for acute thrombosis is rather poor[56] because D-dimer levels can be elevated in other clinical conditions that are associated with enhanced fibrin (malignancy, trauma, increased age, disseminated intravascular coagulation, inflammation, infection, sepsis, postoperative states, and preeclampsia).

Nonsurgical Management

Compression

Compression therapy is the mainstay of treatment for CVI. Because venous hypertension in the upright position and during ambulation is the physiologic cause of the damage in CVI, the first step of treatment should be to reduce the ambulatory venous pressure. Compressive dressings aid venous return by compressing the leg and increasing the interstitial tension. Compression of dilated, engorged, superficial, and intramuscular veins indirectly increases the efficiency of the calf pump mechanism.[43] After the volume stabilizes, the patient can be measured for stockings. Typically, knee-length graduated compression stockings with a pressure of 20 to 30 mm Hg or 30 to 40 mm Hg (at the ankles) are prescribed. Graduated compression providing decreasing pressure from distal to proximal has been the standard of care for both thromboprophylaxis and management of venous and lymphatic disorders. One study showed that a negative graduated compression bandage applied with higher pressures over the calf than the ankle showed significant hemodynamic superiority in patients with severe venous incompetence. Bandages exerting a higher pressure over the calf compared with the ankle were more effective in increasing the venous pump function in patients with venous insufficiency compared with graduated compression.[45] The superior effect of stronger compression over the calf could be explained by more intense squeezing on the venous blood pooled in the calf or a more complete hindrance to venous reflux during walking.[45]

Elevation

When elevation is used for edema control, the extremity is typically elevated above the level of the heart. The patient should lie on a sofa or sit in a recliner to elevate the legs appropriately. The correct duration and frequency of leg elevation should be tailored to the severity of disease. The leg should be elevated when possible, and long periods of standing or sitting with the legs dependent should be avoided. In patients with concomitant arterial disease, lower compression and modified elevation are used to avoid further compromise of arterial inflow.

Intermittent Pneumatic Compression

Intermittent pneumatic compression (IPC) pumps generate short high-pressure waves followed by low-pressure intervals. Single-chamber devices involve a compression force applied uniformly through a chamber vs. multichamber devices, which provide individual or sequential chamber inflation. IPCs can improve calf muscle pump function and venous return and reduce venous stasis.[7] Compression garments should be worn between pumping sessions.

Exercise

Patients with severe CVI are less physically active than age-matched counterparts.[10] The value of exercise in the management of CVI has not been conclusively demonstrated. However, exercising is known to improve microvascular endothelial function and therefore improve venous flow; thus physical activity should still be encouraged in this population.[10] Exercises involving the leg musculature, such as walking, bicycling, or swimming, promote muscle tone in the calf and enhance venous return. Of note, exercise produces variable reductions in venous hypertension. Patients with chronic venous stasis as a result of incompetent deep vein valves generally do not obtain as much reduction in venous pressure as do those in which the primary defect is caused by incompetent perforator valves.[19] Physiatrists should also consider referring patients with CVI to physical therapy, where the goal is to obtain a persistent increase in efficacy of the mechanisms facilitating venous return. In patients with advanced CVI, plantar loading, joint flexibility, and muscular efficiency are negatively affected compared to age-matched controls. During physical therapy, plantar loading should be evaluated and corrected, if necessary; therapy should also address passive and active range of motion at the ankle joint. Calf exercise regimens should be added to improve muscular endurance and possibly restore muscle pump function.[10] Additional studies are needed to assess the translation of improved venous ejection function to improved clinical outcomes of commercially unavailable progressive elastic compression stockings with a negative compression gradient compared to commercially available graduated compression garments.

Lymphedema and Lymphatic Disease

The Lymphatic System

The lymphatic system, composed of lymphatic vessels, lymphoid organs, and lymphatic fluid, is a component of the circulatory system. Its primary function is to transport fluids from the body's intercellular spaces back into the venous system, but it also plays important roles in immune function and in absorbing and moving

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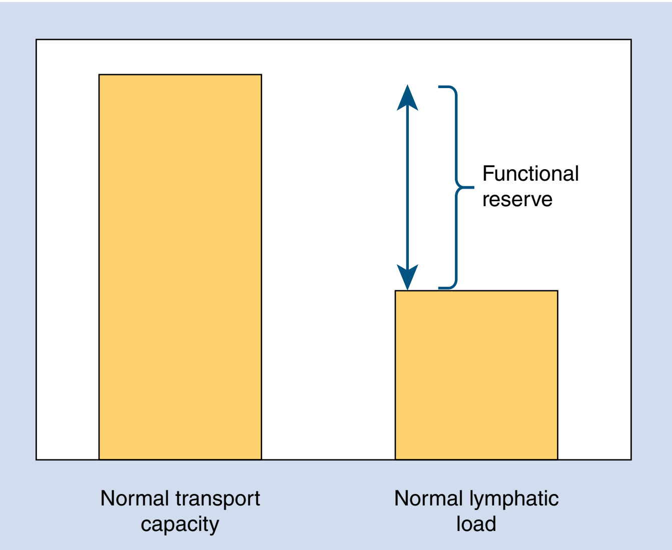
fatty acids from the gastrointestinal system. Embryologically, the lymphatic system develops in close association with the venous system, but unlike the latter, the lymphatic system is not part of a closed circulatory system. Instead, it has a superficial and a deep layer. The superficial layer drains the skin and subcutaneous tissue, while the deep layer is subfascial and is mostly found next to blood vessels, draining the internal organs. The two layers are connected by lymphatic perforating vessels.[74] Interstitial fluid first moves into initial or terminal lymphatics called lymphatic capillaries, which are blind-ended structures. The lymph fluid then drains into collecting lymphatics, which pass through lymph nodes. From there, lymph drains into lymphatic trunks and then to the lymphatic ducts. Of note, lymphatics from the intestinal, hepatic, and lumbar areas drain directly into the cisterna chyli.[65] The thoracic duct, the main collecting system of the lymphatic system, receives drainage from all regions of the body except for the right head and neck, right hemithorax, and right upper extremity. These latter areas drain into the right lymphatic duct.[62] The cisterna chyli and the thoracic duct make up the central lymphatic system. The thoracic duct drains approximately 1 to 2 L of lymph/chyle per day, with 80% of the fluid coming from the liver and intestinal compartments. The remainder of the fluid is from soft tissue, including the extremities. Flow of lymph in the thoracic duct is estimated to be approximately 30 to 190 mL/hour, but it can increase after ingestion of food and water and during abdominal massage. This has implications for diagnosis and treatment in patients who present with central lymphatic abnormalities.[73]

Lymphatic System Physiology and Insufficiency

The amount of lymphatic fluid the system can maximally move is known as the transport capacity. It is equivalent to the maximum lymph time volume. At rest, the lymph time volume is about 10% of the transport capacity. The difference between the transport capacity and the normal lymphatic load, or lymph volume, is called the functional reserve (Fig. 26.8). The functional reserve serves as a safety mechanism, allowing for the lymphatic system to react to increases in lymphatic fluid with a rise in the lymph time volume. Insufficiency of the lymphatic system can occur when the lymphatic load surpasses the transport capacity, resulting in edema.

In dynamic insufficiency, also known as high-volume insufficiency, the lymphatic system and the transport capacity are functioning normally, but the lymphatic load is elevated. Examples include pregnancy, congestive heart failure, venous insufficiency, and immobility. If the dynamic insufficiency is chronic, the lymphatic system can eventually become damaged and fail, resulting in eventual reduction in the transport capacity.

Mechanical insufficiency is another type of lymphatic system insufficiency. Unlike dynamic insufficiency, mechanical insufficiency is due to an abnormally low transport capacity with a normal lymphatic load. This is seen in situations where the lymphatic system is damaged due to lymph node dissections, filariasis, radiation, or innate problems. Dynamic and mechanical insufficiency can also occur together, in a situation known as combined insufficiency. Any of the insufficiency situations described here can result in lymphedema and lymphatic diseases.[74]



• **Fig. 26.8** In a normal functioning lymphatic system, the transport capacity exceeds the lymphatic load.

Lymphedema and Central Lymphatic Diseases

Lymphedema is the abnormal buildup of water and proteins in the subcutaneous tissues due to lymphatic system insufficiency and is estimated to affect more than 200 million people worldwide.[66] It can be divided into primary and secondary forms. Primary lymphedema is due to a developmental abnormality of the lymphatic system and can be congenital or due to hereditary conditions. More common is secondary lymphedema, which is the result of damage to the lymphatics due to infections such as filariasis, obesity, cancer, surgeries such as axillary node dissections, radiation and other iatrogenic causes, inflammation, or trauma. Common types of secondary lymphedema likely encountered by the rehabilitation professional are cancer-related lymphedema and obesity-related lymphedema. Commonly, lymphedema will present with edema in the extremities, but it can also be seen in the head and neck, groin, abdomen, and trunk.

Central lymphatic diseases are due to two main mechanisms: direct trauma to lymphatic vessels and occlusion of the thoracic duct leading to leaky collateral vessels. Common causes of traumatic injury include thoracic, cardiac, and head and neck surgeries. In addition, line placements can also lead to injury. Nontraumatic occlusions are commonly due to malignancies, inflammatory or infectious diseases, primary lymphatic diseases, or are idiopathic.[26,33] Central lymphatic diseases will more commonly present with abdominal distension, abdominal pain, abdominal swelling, or edema of multiple extremities.

Evaluation

The physiatrist's evaluation of the patient with suspected lymphedema should involve gathering a thorough history, specifically taking note of any potential etiologies for lymphedema or other causes of edema. Cardiac, renal, and hepatic histories should be explored in addition to any exposure to risk factors for lymphedema development, such as time spent in a filariasis-endemic area, repeated cellulitis episodes, history of deep vein thromboses, history of trauma, and history of surgeries or other medical treatments that may have damaged the lymphatic system. Patient medications should be examined, as some commonly prescribed drugs, such as gabapentin and amlodipine, are known to cause fluid retention.

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Specific laboratory and imaging/procedures should be ordered to rule out other causes of edema. A comprehensive metabolic panel to assess for renal and hepatic issues and a thyroid panel to assess for thyroid disease should be obtained. An echocardiogram to assess cardiac function and rule out heart failure as a cause of edema should also be completed. Duplex ultrasound can diagnose venous insufficiency, which can cause edema or lead to phlebolymphedema. If there is concern for arterial disease, a pulse volume recording should be performed prior to initiating any lymphedema treatment due to contraindications to compression if the arterial brachial index is significantly low. It is also prudent to obtain imaging of the abdomen and pelvis to look for obstructive etiologies for the patient's edema and if there is suspicion for chylous ascites. If there is concern for chylothorax, imaging of the chest should be done.

On exam, the patient with suspected lymphedema should be examined for any areas of swelling, even those areas that may not have been part of the chief complaint. Primary and secondary forms of lymphedema can present in multiple body areas, as can central lymphatic dysfunction. The latter more commonly presents with fluctuating abdominal distension and discomfort and/or edema of multiple extremities and the head/neck. The edematous areas should be evaluated for pitting and nonpitting swelling, skin changes such as thickening, wounds, or warty growths. Certain fat and lipoma tissue disorders such as lipedema, Dercum disease, and Madelung disease may mimic lymphedema; in these cases, it is especially important to get a detailed clinical history and examine closely the distribution of edema and/or fat tissue deposition. Imaging can be helpful in assessing for fatty cysts vs. edema. The International Society of Lymphology describes four stages of peripheral lymphedema that can be differentiated based on history and physical exam. Stage 0 is the subclinical stage, and at this point the patient is at risk for developing lymphedema. The patient will not present with visible swelling but may have symptoms of heaviness or tingling. In stage 1, lymphedema is reversible, with pitting edema and fluctuations in the volume depending on time of day and in response to treatment. Stage 2 lymphedema is considered the spontaneously irreversible stage because during this stage, the patient has developed lymphostatic fibrosis. In early stage 2, there may still be some pitting edema, but as fibrosis increases and tissue hardens, there is no longer pitting. Stage 3 lymphedema is the most severe stage, characterized by further fibrosis, tissue hardening, and development of complications such as papillomas, hyperkeratosis, and ulcerations (Table 26.1 and Fig. 26.9).[18]

Stemmer sign is a commonly performed physical exam test when assessing for lymphedema. It involves pinching the skin over the second or third proximal toe in the lower extremity or proximal to the metacarpophalangeal joint of the index finger in the upper extremity. Inability to pinch the skin is a positive Stemmer sign and suggests lymphedema. In a study where individuals were tested for Stemmer sign and also underwent lymphoscintigraphy to document lymphatic function, Stemmer sign was found to be 92% sensitive and 57% specific.[25]

Imaging Modalities

While there is debate as to whether lymphedema is a disease that can be diagnosed clinically, the rise of, and increased availability to, imaging techniques to visualize lymphatic dysfunction has allowed for the definitive diagnosis of lymphedema via imaging.[44] Following are some of the most common methods to image the lymphatic system.

TABLE 26.1 International Society of Lymphology Lymphedema Staging System

Clinical Stage	Findings
Stage 0	Latent, in which swelling is not yet visible. There is impaired lymph transport, and there may be changes in subjective symptoms.
Stage 1	Reversible, in which edema will reduce with elevation of the affected body region. Early accumulation of fluid with high protein content. There may be pitting.
Stage 2	Spontaneously irreversible, with edema that rarely reduces with elevation of the affected body region. There is minimal to no pitting as fat and fibrosis accumulate.
Stage 3	Further progression in fat and fibrosis deposition, with trophic skin changes. Warty growths may occur.

From Executive Committee of the International Society of Lymphology. The diagnosis and treatment of peripheral lymphedema: 2020 consensus document of the International Society of Lymphology. *Lymphology*. 2020;53(1):3-19.

• **Fig. 26.9** A 58-year-old female with International Society of Lymphology stage 2 lymphedema of the left upper extremity. She had undergone breast cancer treatment with left-sided mastectomy, axillary lymph node dissection, chemotherapy, and radiation.

Lymphoscintigraphy

Lymphoscintigraphy is considered the gold standard imaging technique for lymphedema diagnosis. It is a nuclear medicine test using radiolabeled technetium sulfur colloid injected intradermally, subcutaneously, and/or subfascially between the web spaces of the hands or feet to visualize lymphatic drainage in the upper or lower extremities. The technique can allow for visualization of both superficial and deep lymphatic pathways. Intradermal

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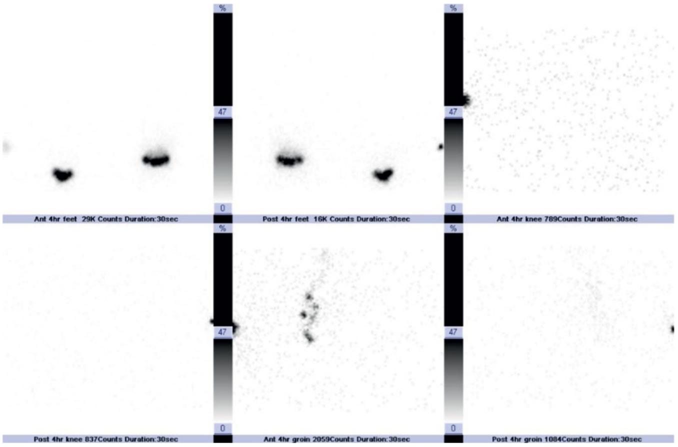

• **Fig. 26.10** Lymphoscintigraphy of the bilateral lower extremities in a 65-year-old female with primary lymphedema. There is absent tracer transit on the left, with no lymphatic channels or nodes visualized at the knees or the groin over the course of the 4-hour study.

injection has been associated with better tracer kinetics than the subcutaneous approach, and it allows for good visibility of the superficial system, whereas the deep lymphatics are better seen via subfascial injections. Lymphoscintigraphy may allow the visualization of normal, reduced, or absent uptake of radiolabeled tracer in the lymph nodes; of whether lymphatic ducts are normal, engorged, or missing; and of presence or absence of dermal backflow.[25,42,44] Unfortunately, there is a lack of a standardized protocol, and sensitivity and specificity vary due to differing techniques. In addition, there is poor image resolution compared to several other lymphatic imaging approaches (Fig. 26.10).

Magnetic Resonance Imaging Lymphangiogram

MRIs are unique in their ability to highlight edema, fat, and contrast agents, and unlike several other modalities, MRIs do not have limitations due to the depth of abnormalities. With MRI, clinicians can visualize extent and location of edema, severity of edema and fatty fibrotic tissue, and lymphatic and venous pathology. MRIs can be used to image both the peripheral and the central lymphatic systems. For imaging of the peripheral lymphatic system, a paramagnetic contrast is injected subcutaneously/intradermally between the web spaces of the hands to image the upper extremities or the feet to image the lower extremities. Lymphatic fluid has a high T2-relaxation time, and a heavily T2-weighted sequence is used to visualize the lymphatic channels and assess for lymphedema (Fig. 26.11).[42] Magnetic resonance imaging lymphangiograms (MRLs) allow for the visualization of abnormalities in the structures of lymph nodes and lymphatic vessels in both primary and secondary pathologies. Due to the time-consuming, technically challenging, and high-cost nature of MRLs, only select institutions may have access to peripheral and/or central imaging capability and radiology interpretation. Basic MRIs are unable to capture images of the central lymphatic system, but more recently, high-resolution imaging techniques have been developed to allow for successful visualization of these deeper structures of the lymphatic system. Nodal dynamic contrast-enhanced MRL (nodal DCE-MRL) is one such approach. In nodal DCE-MRL, ultrasound-guided groin lymph node cannulation is first completed, and MR images are captured with continuous slow injection of diluted contrast agent.[51]

• **Fig. 26.11** Magnetic resonance lymphangiogram (MRL) of the bilateral lower extremities in a 62-year-old female with secondary lymphedema of the right leg after renal transplant complicated by right pelvic lymphocele. The MRL demonstrated severe lymphedema with greater than twice subcutaneous fat accumulation compared to the contralateral side and substantial delay of lymphatic drainage of contrast.

Indocyanine Green Lymphography

Indocyanine green (ICG) technology has been used in medicine for more than 50 years in the assessment of cardiovascular function, hepatic clearance, and retinal angiography. It is also commonly used during sentinel lymph node biopsy procedures during cancer surgical intervention. More recently, it has been applied to assessment of lymphedema and is used to find lymphatic vessels and nodes in real-time microvascular lymphatic surgery. ICG has increased our understanding of lymphatic flow, improving therapeutic approach to manual lymphatic drainage. Unlike lymphoscintigraphy, ICG does not subject patients to radiation. However, it is only able to visualize superficial lymphatics because its penetration depth is 1 to 2 cm below the skin surface.[9]

Ultrasound

Ultrasound is rapidly emerging as a tool for bedside evaluation of the lymphatic system. The technology can demonstrate transition from the normal trilaminar structure of the dermoepidermal complex to the hypoechoic dermal edema seen in early lymphedema.[27,55] This small amount of fluid is not usually enough to

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cause a detectable volumetric variation in the area of concern, giving ultrasound the unique ability to detect early changes in a lymphedematous body part. Additionally, it may allow one to see the dilation of the deep lymphatic collectors in the subcutaneous tissue. Even newer is the use of ultra-high frequency ultrasound (frequencies at about 70 MHz with a resolution of 30 μm) to visualize lymphatic channels during microlymphatic surgeries.[8]

Treatment of Lymphedema

The discussion here will focus on the treatment of peripheral lymphedema because the treatment of central lymphatic issues differs. The treatment of lymphedema should be tailored to each person, accounting for lifestyle as well as medical and psychosocial factors. For successful outcomes, the physiatrist should make a concerted effort to ensure the patient understands the condition and takes an active role in self-care.

Complete Decongestive Therapy

The gold standard of care for lymphedema is complete decongestive therapy (CDT), which is led by a certified lymphedema therapist. CDT involves a combination of manual lymphatic drainage (MLD), compression therapy, decongestive exercises, and skin care. Proper treatment of lymphedema involves both acute management of the edema (phase 1) and continuation with a consistent home maintenance program to prevent the edema from recurring or worsening again (phase 2).

MLD, commonly confused with massage, is the manual movement of lymphatic fluid to increase lymphatic vessel contractions and lymphatic fluid production and to encourage new or specific pathways for lymphatic fluid to flow. It can also provide analgesia, making it a useful technique for pain management, such as in palliative care for painful malignant lymphedema. More recently, the use of ICG to better understand lymphatic anatomy and lymph flow has resulted in improvements in MLD approaches in patients.[35]

Compression therapy involves the use of bandages, bandage alternatives, and garments to increase pressure and improve lymphatic and venous return, to support muscle pumps when active, and to prevent edema from recurring. Short stretch bandages are used in management of lymphedema. Short stretch bandages, unlike long stretch bandages, have low resting pressure and high pressure when one is active and muscles are contracting. Long stretch bandages, such as ACE bandages, are contraindicated in lymphedema management due to their high resting pressure, thus leading to a high risk of injury through the tourniquet effect. Compression garments are available in a variety of forms to fit the affected area (e.g., sleeves, gloves, pantyhose, knee-high stockings, face masks, and toe caps).

Remedial decongestive exercises are movement-based exercises to encourage lymphatic and venous flow. They include active, repetitive, and nonresistive motions of the affected body region. Skin care is also an integral component of treatment given lymphedema's effect on skin integrity and the increased risk for chronic skin disorders, infections, and inflammation in lymphedema.

While CDT is the gold standard, affected persons may be limited in their ability to undergo every aspect of its treatment. For example, patients may not have the ability to undergo compression bandaging of the leg due to their line of work; instead, a bandage alternative such as a Velcro wrap, can be considered for decongestion. Once patients complete CDT, they are then expected to continue their care of lymphedema at home. Commitment to, and consistency with, a home maintenance program with compression, self-MLD, and lifestyle changes cannot be overemphasized. A frequent reason for patients to return to lymphedema therapy for worsening of their lymphedema is lack of adherence to their home program.

Surgical Management

Surgical treatment of lymphedema is typically reserved for lymphedema that has failed nonsurgical management. Surgical treatment options have advanced substantially in the last few decades. The earliest treatment involved removal of subcutaneous and deep fascia followed by coverage with split-thickness grafts in the Charles procedure. More recently, advances in microsurgical techniques have led to lymphovenous anastomosis (LVA) and vascularized lymph node transplant (VLNT). In LVA, lymphatic vessels are connected to venules, allowing lymphatic fluid to bypass the obstruction. One study found that LVA led to a 61% reduction in volume in stage 1 or 2 upper extremity lymphedema.[13] Results are better in upper extremity lymphedema than lower extremity lymphedema.[13,53] In a more recent prospective study examining 100 patients with upper or lower extremity secondary lymphedema who underwent LVA, there were significant improvements in quality of life. Only 35% of upper extremity lymphedema patients had to continue their compression garments, while 60% of lower extremity lymphedema patients continued to need to wear their garments after the procedure. There was also a reduction of more than 50% of mean cellulitis episodes.[53] LVA has also been used in prevention of lymphedema through procedures such as lymphatic microsurgical preventative healing, in which lymphatic-venous anastomoses are created to prevent arm lymphedema at the time of axillary lymph node dissections.[40,54]

VLNT involves transplantation of vascularized lymph nodes to the lymphedema region. It is believed that the lymph node acts as a sponge or induces lymphangiogenesis, and/or lymph vessels themselves act as pumps to move lymphatic fluid. Donor sites include the omental, groin, jejunal mesenteric, and submental regions.[4] Both LVA and VLNT can decrease lymphedema volume, infection rates, and the frequency of garment use in the appropriate settings. LVA and VLNT are best performed in earlier stages of lymphedema, before lymphostatic fibrosis has occurred.[50] When LVA and VLNT are not options, such as in late stages of lymphedema when there is significant fibrosis, suction-assisted protein lipectomy (SAPL) may be considered. SAPL involves removal of excess solid volume, thus reducing limb size rapidly. It has also been shown to decrease the incidence of infections. Because it does not change the physiologic abnormality in the lymphatic system, postoperatively patients must still maintain lifelong compression garment use.[17,24]

Other Treatments

Exercise is a core component of lymphedema management and should be recommended by treating providers for their patients. The majority of the literature on exercise and lymphedema comes from studies on breast cancer–related lymphedema, demonstrating exercise to be safe in patients with lymphedema.[28,36,41,59] In fact, one large study of patients who had been treated for breast cancer found that participation in a guided resistance training program did not increase the risk of lymphedema development.[59] IPC can be utilized as an adjunct to therapy. There are a variety of IPC types on the market, with most offering sequential gradient

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pressure through a device. They simulate manual lymphatic drainage, but they are never a replacement for compression wear.

Treatment of Central Lymphatic Diseases

Treatment of central lymphatic diseases depends on the etiology of the process. For management of chylothorax, noninterventional options include lymphedema therapy, a low-fat diet rich in medium-chain triglycerides, bowel rest, and treatment of an underlying malignancy if that is present. Surgical treatments include thoracic duct embolization, ligation of the thoracic duct, pleurectomy, pleurodesis, and pleuroperitoneal shunts. Management of chylous ascites, protein losing enteropathies, and hepatic lymphorrhea may include interstitial lymphatic embolization. If there is a thoracic duct obstruction and a central lymphatic flow disorder, LVA and thoracic duct venous junction lymphoplasties can be considered, in addition to modified versions of lymphedema therapy. Caution must be exercised when prescribing lymphedema therapy in patients with central lymphatic obstructions because compression and/or lymphatic drainage can potentially worsen patients' symptoms.

Contraindications and Special Considerations

As the treating physiatrist, it is important to recognize when lymphedema therapy is contraindicated or should be modified. Any patient with an acute cellulitis, uncontrolled cardiac insufficiency, uncontrolled renal insufficiency, acute/untreated deep vein thrombosis (DVT), active cancer in the affected area, severe neuropathy in the affected area, and/or severe PAD should avoid lymphedema therapy. In certain cases, therapy can be modified for the patient, and a discussion between the physiatrist and the treating lymphedema therapist is encouraged to develop an appropriate treatment plan. Finally, if a cancer survivor with well-managed lymphedema suddenly presents with acute exacerbation of edema along with pain, further testing to rule out cancer recurrence and DVT must be completed promptly.

Summary

Patients with vascular disease pose a significant challenge for the rehabilitation professional. Arterial, venous, or lymphatic dysfunction may be the primary concern or a critical comorbidity in patients who present to rehabilitation. A detailed vascular history and examination along with selected diagnostic tests should be inherent in the rehabilitation evaluation. When identified early, interventions (including exercise, appropriate compression, positioning, protection, and proper footwear) may ameliorate the need for more aggressive medical and surgical treatment in patients with vascular disease and avoid additional threat to quality of life and function.

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